

VOLODYMYR-VOLYNSKY, Ukraine

WMO: 331770

Lat: 50.8331N

Lon: 24.3436E

Elev: 636

StdP: 14.36

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	5.9	-5.3	4.2	-0.6	1.2	6.0	6.3	27.4	34.5	22.1	31.6	2.5	120	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	86.1	68.9	83.0	67.6	79.9	66.2	70.9	82.2	69.2	79.7	67.6	77.3	5.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	101.5	76.6	65.4	96.0	74.2	63.9	91.1	72.5	35.1	82.8	33.6	79.8	32.3	77.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	15.4	12.0	-6.9	91.0	8.4	2.6	-12.9	92.9	-17.9	94.4	-22.6	95.9	-28.7	97.8		
			-7.3	73.9	8.3	1.6	-13.3	75.0	-18.1	76.0	-22.8	76.9	-28.9	78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	26.7	29.1	36.8	48.6	57.5	63.3	66.9	65.4	56.7	47.5	38.6	30.0
	DBStd	16.19	10.36	10.52	8.55	7.49	6.73	5.80	5.41	5.53	6.47	7.53	8.40	9.54
	HDD50	2972	722	585	413	114	14	0	0	0	14	140	349	621
	HDD65	6693	1187	1005	875	494	246	102	43	62	258	544	791	1086
	CDD50	2006	0	0	3	71	246	399	524	478	216	61	8	0
	CDD65	252	0	0	0	1	14	50	102	75	10	0	0	0
	CDH74	2522	0	0	0	21	193	480	952	758	116	2	0	0
	CDH80	636	0	0	0	2	27	94	269	227	17	0	0	0
Wind	WSAvg	4.9	5.7	5.6	6.1	5.4	4.7	4.3	3.9	3.8	4.1	4.7	5.3	5.4
Precipitation	PrecAvg	25.5	1.5	1.4	1.5	1.8	2.7	3.1	3.7	2.6	2.4	1.8	1.6	1.5
	PrecMax	30.6	3.3	2.8	3.2	3.4	5.4	6.7	8.1	8.8	5.2	5.0	3.1	2.7
	PrecMin	19.8	0.2	0.3	0.3	0.6	0.8	0.8	0.6	0.1	0.4	0.2	0.1	0.6
	PrecStd	3.0	0.8	0.6	0.7	0.7	1.1	1.6	2.0	1.7	1.4	1.2	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	52.6	62.8	76.6	83.5	86.4	89.7	90.8	82.1	71.9	61.5	49.5
		MCWB	43.6	46.2	50.8	59.5	65.1	69.3	70.7	69.7	65.6	61.1	53.6	46.0
	2%	DB	43.9	47.4	56.7	70.8	79.1	83.0	86.3	85.9	77.4	67.1	56.5	45.9
		MCWB	40.8	43.1	47.2	55.9	63.4	67.6	69.5	68.6	63.8	58.1	51.4	43.1
	5%	DB	40.9	43.9	52.6	65.8	75.4	79.6	83.0	81.9	73.1	63.0	53.0	43.0
		MCWB	38.5	40.6	45.1	53.6	62.1	65.9	68.3	67.4	61.8	56.3	49.4	40.8
	10%	DB	38.5	41.1	48.6	61.9	71.3	76.3	79.5	77.9	69.0	59.0	49.8	40.6
		MCWB	36.5	38.6	43.4	51.5	59.7	64.5	67.0	65.4	59.8	53.6	47.1	38.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.9	46.7	51.6	61.0	67.3	71.5	73.7	72.9	68.1	62.1	54.4	46.5
		MCDB	46.5	51.1	61.3	74.9	78.9	82.2	85.2	85.2	79.8	69.3	59.2	49.0
	2%	WB	41.0	43.5	48.4	57.6	64.8	69.2	71.4	70.3	65.1	59.3	52.1	43.4
		MCDB	43.4	46.9	55.0	68.0	76.0	79.9	82.8	81.9	74.2	65.7	55.8	45.5
	5%	WB	38.7	41.0	46.0	55.0	63.0	67.3	69.7	68.4	62.8	56.8	49.7	41.0
		MCDB	40.6	43.3	51.4	63.4	73.7	77.1	80.1	78.9	70.6	62.1	52.9	42.8
	10%	WB	36.6	38.5	43.7	52.8	60.9	65.5	67.9	66.7	60.8	54.2	47.1	38.8
		MCDB	38.3	41.1	48.1	59.9	69.3	73.9	77.4	75.8	67.8	58.4	49.6	40.5
Mean Daily Temperature Range	5% DB	MDBR	8.2	10.0	12.8	17.4	19.0	18.4	18.7	19.8	17.1	14.2	9.3	7.9
		MCDBR	8.6	12.1	20.3	24.7	26.2	24.3	25.4	26.7	25.1	20.7	15.4	9.3
		MCWBR	7.2	8.9	12.5	12.4	12.8	11.7	11.2	11.8	13.2	12.7	10.6	7.6
	5% WB	MCDBR	7.9	11.5	18.1	22.0	23.2	21.7	22.1	22.6	22.1	19.5	13.2	8.9
		MCWBR	6.9	8.9	11.8	11.7	12.1	11.4	10.7	11.0	12.6	12.6	9.6	7.7
Clear-Sky Solar Irradiance	taub	0.313	0.342	0.380	0.431	0.419	0.415	0.443	0.441	0.404	0.366	0.349	0.310	
	taud	2.221	2.182	2.190	2.126	2.223	2.260	2.198	2.199	2.259	2.340	2.299	2.274	
	Ebn at Noon	218	243	256	255	266	267	257	250	246	234	204	201	
	Edh at Noon	23	31	37	44	42	41	43	41	35	27	22	19	
All-Sky Solar Radiation	RadAvg	264	518	889	1325	1662	1789	1700	1527	1041	618	274	185	
	RadStd	43	76	118	137	161	130	147	108	152	102	30	32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.97	N/A	N/A	+1.39	N/A	N/A	N/A	-308	+157	+72
(47)	Regional (14 neighbors)	+1.03	N/A	N/A	+1.62	+0.86	+0.76	N/A	-337	+184	+76

Nomenclature: See separate page